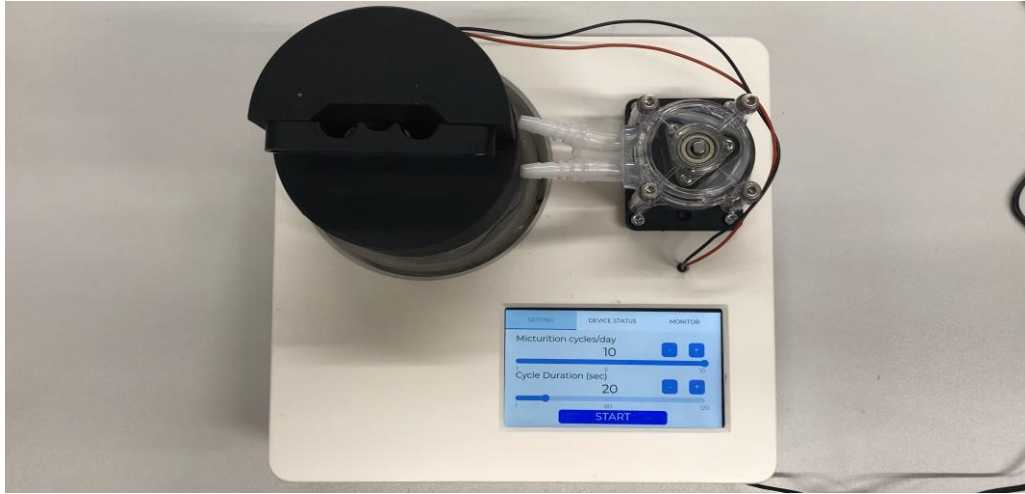
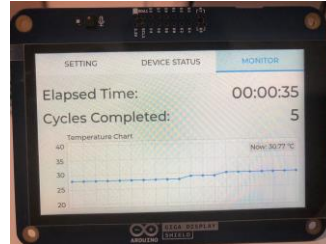


Automated Test Bench Development (Grade: 1.3/Sehr gut)

User Interface and Control Software for an Automated Test Bench for the Analysis of Biofilm Formation



TOUCHSCREEN GUI (LVGL)



OVERVIEW

Designed and implemented an automated test bench for long-term evaluation of biofilm formation on implantable medical devices. The platform combines embedded control, touchscreen interaction, environmental monitoring, and automated micturition-cycle simulation in one integrated system to support repeatable durability testing.

MY CONTRIBUTIONS

- Developed modular control software in C++
- Designed touchscreen GUI using Arduino GIGA + LVGL
- Implemented automated micturition cycle scheduling
- Integrated temperature and environmental sensors
- Controlled pump and UV sterilization subsystems
- Built reliable logic for long-term continuous operation
- Structured system for future expansion and validation

RESULT

- Enabled 6+ months continuous operation
- Stable automated execution of repeated fluid cycles
- Reliable sensor monitoring and live visualization
- Improved repeatability for implant durability testing
- Scalable platform for future validation studies